Towards a FAIR Framework for Earthquake catalogues in Aotearoa New Zealand

A Community White Paper

Making New Zealand's earthquake catalogues Findable, Accessible, Interoperable, and Reusable

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Executive Summary

Earthquake catalogues underpin seismology, monitoring, engineering, and risk management. Aotearoa New Zealand produces many of them (the GeoNet national catalogue alongside regional, volcano-specific, temporary-deployment, historical, and derived catalogues), but they are fragmented: built to different standards, documented inconsistently, and rarely interoperable. The 2022 National Seismic Hazard Model had to re-standardise magnitudes across sources before it could use them (Gerstenberger et al., 2024; Christophersen et al., 2024); high-value research catalogues such as the 86,579-event Taupō Volcanic Zone catalogue (Illsley-Kemp and Mestel, 2025) sit outside any common system; and location quality across the national catalogue varies markedly with station coverage (Warren-Smith et al., 2025). Combining catalogues today is a manual, error-prone task. A companion registry-style inventory (1960–2026) documents 94 of them, spanning national, regional, temporary, historical, and derived types.

Two national workshops (2023 and 2024) brought together researchers, agencies, and industry to identify needs and shape a practical framework guided by the FAIR principles (Findable, Accessible, Interoperable, Reusable). This white paper sets out the current landscape, the community-identified gaps, and a proposed national framework: a "catalogue of catalogues" that *complements rather than replaces* GeoNet. The framework makes catalogues findable through a common submission profile and provides FAIR services (quality assessment, merge and conflation, analytics, versioning, and programmatic access) for the catalogues it coordinates. It is accompanied by proposed governance, a roadmap, and an indicative resourcing envelope. A working prototype demonstrates the technical feasibility of the core workflows, from ingestion and merging through analytics to multi-format export. We ask New Zealand's earth-science community, its agencies, and its research-infrastructure funders to review and refine the standards proposed here and to help establish a sustained national capability.

Key Recommendations

1. Adopt a minimum-attributes *submission profile* and a controlled parameter lexicon, aligned with the international QuakeML and FDSN seismic-data standards, so every catalogue is described consistently (distinguishing required from recommended fields).
2. Establish the openly versioned "catalogue of catalogues" as the low-barrier, near-term community deliverable, beginning with discovery: a searchable index of submission profiles that records which catalogues exist and where to find them.
3. Build out the FAIR services incrementally (a programmatic API, cross-agency event-identifier mapping, transparent merge/conflation, configurable quality scoring, and analytics) for the catalogues it coordinates.
4. Require persistent identifiers (DOIs), explicit versioning, and machine-readable changelogs for DOI-bearing catalogue releases, so any cited catalogue version is reproducible.
5. Represent uncertainty explicitly: report uncertainty values where available, and where they are absent record that absence or a qualitative confidence class, so that no record implies more precision than the data support.
6. Establish formal governance (a Steering Committee with substantive Māori representation, and a Data Stewardship Team) that operationalises both FAIR and CARE principles, underpinned by sustained multi-year funding.

Keywords: earthquake catalogues; FAIR principles; CARE principles; seismology; research data infrastructure; data governance; Aotearoa New Zealand.

Contents

1 Introduction: The Problem and the Opportunity

Earthquake catalogues are the cornerstone of seismology and the shared evidence base for hazard models, engineering design, rapid response, and fundamental research (Woessner et al., 2010). Their value depends on being trustworthy, comparable, and reusable.

In Aotearoa New Zealand that evidence base is fragmented. GeoNet curates the authoritative national catalogue, with broad coverage and open access through web and programmatic services (GeoNet, 2025a,c,d); but its location quality and completeness vary markedly across the country with station density (Warren-Smith et al., 2025), and metadata conventions have evolved over time. Alongside it sit many high-value catalogues that are effectively invisible to the national system:

- regional and volcano-specific catalogues built with dense local networks or specialised methods, for example the consistent, high-precision Taupō Volcanic Zone catalogue of 86,579 events (Illsley-Kemp and Mestel, 2025), central Alpine Fault microseismicity from the SAMBA borehole array (Michailos et al., 2019), and nested-tomography relocations of Hikurangi subduction seismicity (Aziz Zanjani et al., 2021);
- temporary aftershock and ocean-bottom deployments with lower magnitude thresholds, such as the dense relocations of the 2016 Kaikōura aftershock sequence (Lanza et al., 2019) and the offshore HOBITSS ocean-bottom catalogue on the northern Hikurangi margin (Yarce et al., 2019);
- historical and pre-instrumental compilations from written and oral records, including Eiby's annotated list of New Zealand earthquakes from 1460 (Eiby, 1968), the *Atlas of Isoseismal Maps of New Zealand Earthquakes* (Downes, 1995), and revised magnitudes for 1901–1993 (Dowrick and Rhoades, 1998); and
- derived and synthetic catalogues built for hazard and risk modelling, from the magnitude-homogenised catalogue prepared for the 2022 National Seismic Hazard Model (Christophersen et al., 2024) to multi-hundred-thousand-year RSQSim synthetic catalogues (Shaw, 2021).

These are not isolated exceptions. A companion inventory covering 1960–2026 identifies 94 such catalogues, spanning regional and volcano-seismic networks, temporary onshore and offshore deployments, aftershock responses, historical compilations, and hazard-model datasets; the true number is higher still. Temporary deployments (onshore and offshore combined) are the largest group, yet they remain among the least discoverable through the national system (see Data and Code Availability).

Because these catalogues use different formats, identifiers, magnitude scales, and metadata, combining them is manual and error-prone. The 2022 National Seismic Hazard Model illustrates the cost: a flagship national product required a dedicated effort to standardise magnitudes across sources before they could be used (Gerstenberger et al., 2024; Christophersen et al., 2024). The consequences of the status quo are lost researcher time, reduced reproducibility, and less transparent uncertainty in decisions that can affect public safety and major infrastructure.

The opportunity is to add a thin coordinating layer (a "catalogue of catalogues") that makes these datasets findable, comparable, and citable. In practice it combines a common submission profile that makes catalogues findable with a set of richer FAIR services (comparison, merging, and citable versions) for the catalogues it coordinates. Internationally, initiatives such as the European Plate Observing System (EPOS) Seismology service show the value of standardised, FAIR data services and coordinated infrastructure (EPOS, 2025b,a). This paper synthesises outcomes from two national workshops and intersession working-group activity to propose such a framework, tailored to New Zealand and governed under both FAIR and CARE principles.

This paper has three aims: (i) to characterise the landscape of New Zealand's earthquake catalogues and benchmark it against mature international systems; (ii) to propose a common set of minimum attributes for describing and creating catalogues; and (iii) to set out a gap analysis and near-term community steps toward a federated, harmonised national catalogue capability for earthquake science and engineering.

Terminology in this paper is used as follows. The *framework* (the "catalogue of catalogues") is a single coordinating layer: every catalogue is described by a common submission profile (Section 4.2) so that it is discoverable, and the framework provides FAIR services (validation, cross-catalogue identifiers, merging, analytics, exports, and DOI-bearing releases). A catalogue's data may remain externally held and linked, or be ingested in full so that it can use the services that operate on event-level data.

The framework is shaped by concrete user needs identified at the workshops: gaining an overview of which catalogues exist and where to find them; locating catalogues for a specific region; merging and comparing catalogues; and identifying which catalogues meet the requirements of a given analysis (for example, those that include phase picks, not just hypocentres). Figure 1 contrasts the current and proposed states.

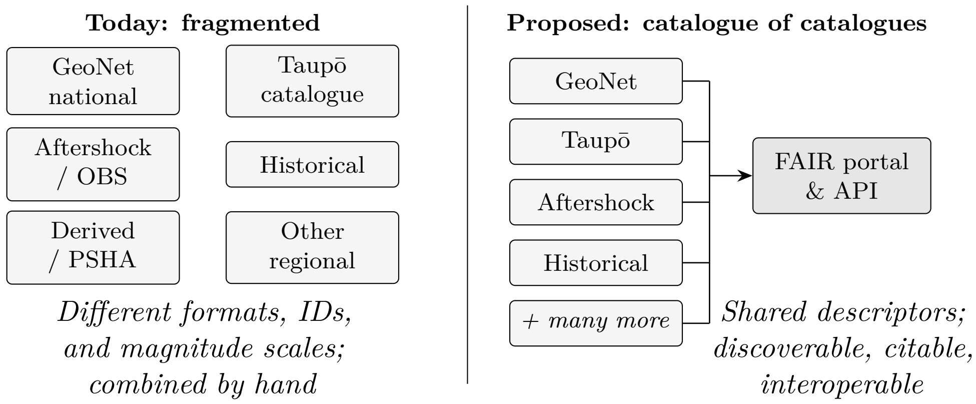

Figure 1: The problem and the opportunity (representative types shown; the full inventory lists 94 catalogues). Today, New Zealand's earthquake catalogues are developed independently and combined manually. The proposed framework adds a coordinating layer that makes them interoperable through shared standards, a common portal, and a programmatic API.

2 The Current Landscape and International Benchmark

New Zealand's earthquake data ecosystem comprises several catalogue types that serve different purposes. The **national operational catalogue**, maintained through GeoNet by Earth Sciences New Zealand (ESNZ; formerly GNS Science), provides consistent coverage and open access since the 1930s (GeoNet, 2025a,c; GNS Science, 2020); its gaps are evolving metadata conventions, mixed magnitude types, spatially variable location quality (Warren-Smith et al., 2025), and limited exposure of detailed processing fields. Around it sit regional and volcano-specific catalogues offering high-resolution detail (Illsley-Kemp and Mestel, 2025), valuable but poorly discoverable temporary deployments, and historical, derived, and synthetic compilations that are difficult to reuse without formal versioning, identifiers, and processing documentation (Gerstenberger et al., 2024). In short, the ecosystem's strengths lie in local precision and national consistency, while its gaps lie in completeness through time and space, processing consistency, magnitude and depth scales, metadata, transparency, and access to legacy and temporary data.

This ecosystem is large and heterogeneous. Table 1 illustrates the main catalogue types contributed by the community, with representative examples and their characteristic strengths and limitations.

Table 1: Representative earthquake-catalogue types in Aotearoa New Zealand, with examples and characteristic strengths and limitations (from the community workshops).

Type	Examples	Strengths / limitations
Volcano-seismic / microseismic	Taupō, Whakaari, Ruapehu, geothermal fields	Precise depths, uniform local methods; limited spatial scope, weak interoperability
National operational	GeoNet rapid and reviewed, GeoNet centroid moment tensors (CMT), strong-motion datasets	Consistency, continuity, accessibility, manual review; evolving conventions, mixed magnitude types
Local / regional deployments	Alpine Fault OBS, Hikurangi (e.g. HOBITSS), Kaikōura/Cook Strait aftershocks, TVZ experiments (SAMBA, COSA, PTANGO)	High-quality velocity models, precise locations and depths; short duration, limited discoverability
Source-mechanism	Ristau moment tensors, Global Centroid Moment Tensor (GCMT), finite-fault and stress-drop inversions	Kinematic and stress information for hazard and tectonics; standalone, weakly integrated
Historical / pre-instrumental	Pre-1940 compilations, oral and written records, NSHM inputs	Extend the record by centuries; large uncertainties, poor depth/location, few small events
Slow-slip, palaeoseismic, geodetic	Slow-slip detections, palaeoseismic records, offshore geodetic detections	Unique long-timescale and aseismic information; sparse coverage, imprecise timing
Research / method-specific	Template-matching, machine-learning, relocation, and waveform-inversion catalogues	Gains in completeness and precision, documented methods; bespoke formats, variable access
Derived / international	NSHM-derived snapshots; ISC, ComCat, EMSC combined catalogues	Broad or standardised coverage; not NZ-specific, variable completeness

Mature international systems provide useful reference points: the USGS Advanced National Seismic System Comprehensive Earthquake Catalog (ComCat) federates the regional and national networks of the ANSS (USGS, 2025); the Swiss Seismological Service couples a dense network with cross-border data (Diehl et al., 2025); the European-Mediterranean Seismological Centre (EMSC) and the International Seismological Centre (ISC) aggregate across many agencies (European-Mediterranean Seismological Centre, 2025b; International Seismological Centre, 2025a); and EPOS provides standardised FAIR services across Europe (EPOS, 2025b). These systems share common standards, notably the International Federation of Digital Seismograph Networks (FDSN) web services and the QuakeML data format, which New Zealand already supports. Appendix A summarises each. The comparison is a qualitative synthesis from public documentation and expert judgement, not a scored audit; the New Zealand column reflects the authors' assessment across New Zealand's many catalogues, whereas the comparators are single national or federated systems. Table 2 compares these systems across the rows shown; five recurring priorities stand out: (1) consistent representation of uncertainty or confidence; (2) formal version control with Digital Object Identifiers (DOIs) and changelogs for citable releases; (3) a cross-agency event-identifier mapping service; (4) integration of historical and pre-instrumental data; and (5) formal governance with dedicated funding. These gaps motivate the framework components and Key Recommendations that follow.

Table 2: Comparison of NZ and International Earthquake Catalogue Capabilities. A qualitative synthesis from public documentation and expert judgement, not a scored audit; the third column cites SED for national-service capability and EPOS for the pan-European federation layer, with pan-European capabilities prefixed "EPOS:" where shown.

Feature	**New Zealand**	**USGS ComCat**	**Switzerland (SED) / EPOS**	**Gaps & Priorities**
Core parameters	Hypocentre (origin time, location, depth), magnitude, phase picks/arrivals, station counts; QuakeML carries location/magnitude uncertainties and multiple magnitudes (GeoNet, 2025a; GNS Science, 2020)	Hypocenter, magnitudes, phase data, moment tensors, macroseismic data (USGS, 2025)	Historical + instrumental events, homogenised local-magnitude scale (M_{Lhc}) with M_w for notable events, uncertainties, intensities (Diehl et al., 2025)	Lacks full uncertainty propagation for all events
Metadata & provenance	GeoNet documents core processing; provenance incomplete or inconsistent across the wider NZ catalogue ecosystem	Maintains standards, data dictionaries, method documentation (USGS, 2025)	Documented processing, 3D velocity models, homogenised ECOS database (Diehl et al., 2025; Fäh et al., 2011)	Need consistent, centralised metadata policies

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Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Switzerland (SED) / EPOS	Gaps & Priorities
Versioning & review	Rapid + review workflows; some reprocessing	Versioned updates, multiple origins per event, revisions over time (USGS, 2025)	Continuous access to evolving datasets and harmonisation (EPOS); static snapshots such as ECOS-09 (SED); no per-version catalogue DOIs/changelogs (Diehl et al., 2025; EPOS, 2025b)	Formalise version control (DOIs, changelogs)
Access & APIs	Quake Search and WFS (CSV, GeoJSON, KML, GML), FDSN (QuakeML), GeoNet API (GeoNet, 2025c,a)	FDSN API, GeoJSON/CSV feeds, search interfaces (U.S. Geological Survey, 2025)	EPOS: integrated access to waveforms and parametric data via standards (EPOS, 2025b)	Enhance federated discovery, cross-catalogue queries

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Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Switzerland (SED) / EPOS	Gaps & Priorities
Latency / solution status	GeoNet automatic solutions within minutes, then manually reviewed; periodic reprocessing	Real-time GeoJSON feeds; preferred solution revised as reviews land (U.S. Geological Survey, 2025)	SED automatic then reviewed; EMSC near-real-time; ISC reviewed bulletin issued ∼24 months later (International Seismological Centre, 2025b; Diehl et al., 2025; European-Mediterranean Seismological Centre, 2025b)	Expose solution status (automatic vs reviewed) and update latency uniformly, per record
Historical data	Some pre-instrumental events with greater uncertainty (GNS Science, 2020)	Includes large historic events, variable completeness (USGS, 2025)	Centuries of data, macroseismic fields, uncertainties (Diehl et al., 2025)	Strengthen integration, metadata, uncertainty modelling
Coverage & completeness (M_c)	$M_c \approx M_L$ 1–2 in well-monitored regions; varies spatially and through time (Warren-Smith et al., 2025; GNS Science, 2020)	$\sim M4.5+$ globally (common analysis threshold); $\sim M2.5$ in well-monitored US regions (USGS, 2025)	M_c 1.0–1.5 via a dense network; record from 1879 (historical to 250 AD) (Diehl et al., 2025; Fäh et al., 2011)	Quantify and report completeness (M_c) and coverage consistently across NZ catalogues

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Table 2 – *continued*

Feature	**New Zealand**	**USGS ComCat**	**Switzerland (SED) / EPOS**	**Gaps & Priorities**
Uncertainty & quality	GeoNet reports location/magnitude uncertainties and quality metrics (RMS, azimuthal gap, station/phase counts) via QuakeML; uneven across other NZ catalogues (Warren-Smith et al., 2025; GeoNet, 2025c)	Quality metrics: RMS, station count, azimuthal gap (USGS, 2025)	Emphasises uncertainty, solution quality, EPOS standards (Diehl et al., 2025; EPOS, 2025b)	Universal uncertainty reporting needed
Multi-catalogue linking	Mostly separate; limited cross-catalogue mapping	Aggregates many networks, links multiple origins per event (USGS, 2025)	EPOS: harmonised catalogues across Europe and cross-border exchange; EMSC/ISC reconcile bulletins across many agencies (Appendix A) (EPOS, 2025b)	Build Event ID mapping service, federated portal
Special products	Focal mechanisms, moment tensors, slow slip as standalone datasets	Hosts moment tensors, additional products tied to events (USGS, 2025)	EPOS: integrates fault, slip, and seismological products via standardised services (EPOS, 2025b)	Better integrate special-purpose catalogues

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Table 2 – *continued*

Feature	**New Zealand**	**USGS ComCat**	**Switzerland (SED) / EPOS**	**Gaps & Priorities**
Standards compliance	Supports FDSN/QuakeML; not all catalogues fully adhere (GeoNet, 2025c)	FDSN and QuakeML in APIs and exports (U.S. Geological Survey, 2025)	Implements FDSN web services (event, station, dataselect); QuakeML and StationXML (Diehl et al., 2025)	Ensure full compliance across all catalogues
Governance & sustainability	ESNZ/GeoNet or individual researchers; less formal structure	Formal governance, funding, institutional roles (USGS/ANSS/NEIC) (USGS, 2025)	Robust institutional backing, cross-national coordination (Diehl et al., 2025)	Establish formal governance, roles, funding

3 A Community-Driven Process

The framework proposed in the following sections was shaped through community engagement. Two pre-conference workshops at the Geoscience Society of New Zealand annual meetings provided the primary engagement: a 2023 workshop framed needs and objectives, and a 2024 workshop reviewed progress and refined priorities (Geoscience Society of New Zealand, 2023, 2024). Participants spanned university researchers, GeoNet and Earth Sciences New Zealand staff, government agencies, and engineering and insurance users. Discussion was structured around eight discussion tables (covering minimum catalogue attributes, metadata and documentation, access and versioning, governance, and community priorities), whose outcomes are synthesised as nine cross-cutting themes in Appendix C. Throughout this period, a working group met regularly to draft standards, test integration pathways, and build the prototype portal.

The workshops were designed as scoping and prioritisation exercises rather than as a statistically representative survey or a formal consensus vote. This paper therefore treats their outputs as community-informed evidence for needs and design direction; the proposed standards, governance model, and implementation roadmap are offered for review, calibration, and partner agreement.

3.1 Prototype Portal: Proof of Concept

To ground the framework in a working system, the working group built a functional prototype portal (*Catalogue of Catalogues*, or CofC) for the framework services of Section 4 (ingestion, merge, export, and analytics), using representative New Zealand datasets. Its design drew on the 2023 and 2024 workshops (Geoscience Society of New Zealand, 2023, 2024), and it was first presented, as a conference talk by the authors rather than a workshop with a published page, at the 2025 Geoscience Society of New Zealand annual meeting, where further community feedback informed its development. The prototype is open-source (MIT licence; see Data and Code Availability). It is a proof of concept that exercises the workflows below functionally, establishing their technical feasibility; the national framework of Section 4 builds on that demonstration. The prototype implements:

- **Ingestion** of CSV/tab-delimited text (with automatic delimiter and date detection), JSON/GeoJSON variants, and QuakeML 1.2, with an alias-based field-mapping interface and direct, de-duplicating sync from GeoNet's FDSN Event service;
- **Search and visualisation**: spatial (bounding box, polygon, radius), time, magnitude, and depth filtering on an interactive map with event clustering, uncertainty ellipses, focal mechanisms, and station coverage;
- **Merge and conflation** with five configurable strategies, conflict detection and logging, lineage tracking, and a pre-commit preview (Section 4.4);
- **Analytics**: b-value, completeness magnitude (M_c), moment release, declustering, and magnitude-frequency comparison, using standard methods (Aki, 1965; Wiemer and Wyss, 2000; Gardner and Knopoff, 1974); and
- **Multi-format export** (CSV, GeoJSON, KML, QuakeML 1.2), including version-specific exports.

Community feedback from the workshops and the 2025 conference presentation informed refinements to the metadata schema, merge options, and quality-grading approach described in Sections 4.4 and 4.5. The pilot phase (Section 5) will validate the prototype on named test catalogues, reporting event counts, accepted and rejected records, merge precision and recall, identifier-matching error rates on dense aftershock sequences, failure cases, and runtime and storage behaviour, with reproduction instructions.

4 The Proposed FAIR Framework

The framework addresses the end-to-end lifecycle of catalogue data; Figure 2 summarises how its components fit together. This section describes the framework conceptually. Implementation-level detail (API endpoints, access-control roles, field-level representations, and validation thresholds) is collected in Appendix B so that this section stays readable for a general audience.

Unless otherwise stated, the components in this section are proposed design requirements. The discovery-level requirements apply to the submission profile that describes every catalogue and are intentionally lightweight; the service-level requirements apply to catalogues ingested in full, and especially to DOI-bearing releases, and are correspondingly stricter.

The framework is a single coordinating layer, the "catalogue of catalogues". Its common front door is a short submission profile (Section 4.2): each catalogue is described by the profile, recording what it contains and how to obtain it, so that the catalogue is discoverable without necessarily transferring the underlying data. This discovery function is cheap to contribute to and is the immediate community deliverable, answering *which catalogues exist, what they contain, and where to find them.* Building on it, the framework provides the richer FAIR services (programmatic access, cross-catalogue event-identifier mapping, merge and conflation, quality scoring, analytics, and versioned DOIs) for catalogues that are ingested in full, answering *compare, merge, analyse, and cite.* A catalogue can therefore be described by a profile that links to wherever it lives, or opt in to full ingestion for the services that operate on event-level data. Discovery should be established first and at low cost; the full services are demonstrated by the prototype (Section 3.1) and would be built out across the roadmap (Section 5). At the time of writing the index is a design proposal: it has not yet been populated, and the New Zealand catalogues named in this paper are motivating examples rather than live entries. Standing up an initial index with an agreed first set of catalogues, via the submission profile, is the near-term deliverable (Section 5). The components below describe both the discovery profile and these services.

Table 3 summarises the framework's two functions, discovery and curation services, before the detailed architecture in Figure 2.

Table 3: Discovery and curation functions of the framework.

Function	**Stores**	**Provides**
Discovery	Catalogue-level descriptors (submission profiles) and links to external data sources	Findability, contact information, broad content flags, and transparent version history of entries
Curation services	Fully ingested event-level catalogue data and provenance	Validation, event-ID mapping, merge/conflation, analytics, exports, and DOI-bearing releases

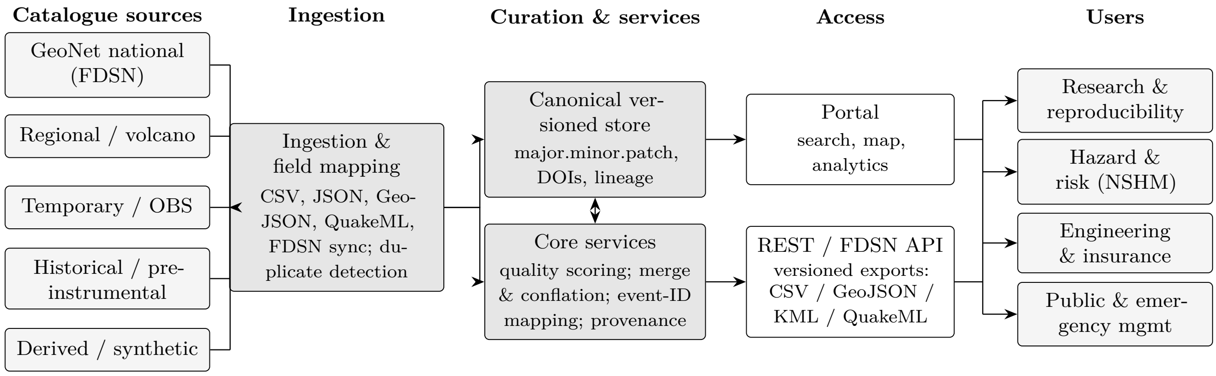

Figure 2: End-to-end architecture of the proposed framework. Heterogeneous catalogue sources selected for full ingestion are mapped to a canonical, versioned store; curation services (quality scoring, merge and conflation, event-identifier mapping, and provenance) feed a portal and a programmatic API that serve research, hazard, engineering, and public users. Capabilities identified as implemented in Section 3.1 are exercised functionally by the prototype. The lightweight discovery entry point (the catalogue submission profile, Table 4) is not shown as a separate store here: every catalogue is described by a profile and linked, whether or not it is ingested in full.

4.1 Metadata Standards and a Shared Lexicon

A core metadata profile aligned with QuakeML and FDSN conventions captures identifiers, origin parameters with uncertainties, magnitude types with errors, detection and quality metrics, methodology descriptors, event classification, and catalogue-level metadata (ETH Zürich Seismology / QuakeML Working Group, 2015; GeoNet, 2025c; U.S. Geological Survey, 2025). Required versus recommended fields are distinguished to balance utility against effort; the full field hierarchy is given in Table 8 (Appendix B). The field-level mapping to QuakeML elements, including how constructs without a native QuakeML element (the composite quality score, cross-catalogue merge lineage, qualitative confidence classes, and macroseismic intensity) are carried via QuakeML's extension and comment mechanisms, is given in Table 10 (Appendix B). A controlled vocabulary standardises magnitude types, event classifications, phase names and quality codes, units and formats, and dataset naming, improving interpretability and filtering across catalogues (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016). An illustrative subset of this lexicon is given in Table 9 (Appendix B).

4.2 Catalogue Submission Profile (Minimum Attributes)

Every catalogue in the framework is described by a concise *submission profile*, the minimum set of attributes prioritised at the 2024 community workshop. The profile records, consistently and searchably, *what* a catalogue contains and *how* to obtain it, so that users can discover catalogues and make a preliminary assessment of relevance before reading the catalogue's full documentation. Required fields are kept to a small core to lower the barrier to contribution, and controlled drop-downs are preferred over free text wherever possible to keep entries searchable and consistent. The profile complements the event-level metadata profile (Table 8, Appendix B): the profile describes a catalogue *as a whole*, whereas the metadata profile describes the recommended internal structure of an ingested catalogue. Table 4 lists the fields.

Table 4: Catalogue submission profile: minimum attributes for a catalogue-of-catalogues entry, as prioritised at the 2024 workshop. Controlled options are shown in this style; fuller detail is carried by the linked dataset/DOI where one exists.

Field	**Status**	**Options / notes**
Identification and contact		
Catalogue name	Required	Short, human-readable title
Contact person(s)	Required	At least one; name and email, multiple separated by semicolons
Data source	Recommended	Origin of the catalogue (e.g. parent catalogue, instruments, records)
Brief description	Required	Why the catalogue was produced and its focus; placed at the end of the form, with a character limit
Additional notes	Optional	Free-text notes/keywords; placed last
Coverage		
Start date	Required	
End date	Optional	Or tick continuous/ongoing
Bounding box	Required	Latitude/longitude (N/S/E/W)
Depth range	Recommended	Minimum and maximum depth
Region of interest	Recommended	Descriptive focus or extent
Magnitude info	Optional	Magnitude range and/or magnitude of completeness, not required (many historical/intensity catalogues lack instrumental magnitudes)
Content present (tick all that apply)		
Content flags	Recommended	hypocentral locations; relative relocations; phase arrival-time picks; amplitude picks; fault-geometry (non-point-source); historical observations (felt reports / macroseismic intensities); quality information; other
Classification (controlled drop-downs)		
Catalogue type	Recommended	background / aftershock / induced / historical / compilation / large-events-only
Focus	Recommended	geographical / parameter-testing / particular-event-type
Detection method	Recommended	manual / automatic / template-matching / machine-learning / felt-reports / other
Review status	Recommended	non-reviewed / partially reviewed / reviewed
Availability		
Data availability	Required	available (DOI) / available on request / not available

Consistent with this scope, the catalogue entry stays high-level and is a discovery record, not a certification of scientific adequacy. Deeper detail is carried by the catalogue's own dataset or publication and referenced by DOI where one exists, rather than duplicated in the entry. This includes velocity models, station and pick information, processing provenance and parameters, waveform and raw-data links, focal-mechanism details, input-catalogue lineage, links to GeoNet event IDs, and any associated-publication abstract. A graphical, map-based discovery interface (polygon selection and faceted filtering) and contributor self-service editing are planned as future enhancements; in the interim, catalogue entries are openly searchable and exportable, while catalogues ingested in full additionally gain the services described below.

4.3 Interoperability: Programmatic Access and Cross-Catalogue Identifiers

A unified API provides machine-readable access across catalogues, returning CSV, GeoJSON, and QuakeML, with bulk downloads of versioned datasets to support reproducible research; it builds on existing GeoNet interfaces and standards (GeoNet, 2025d,c; ETH Zürich Seismology / QuakeML Working Group, 2015). The endpoint design is given in Appendix B.

A persistent interoperability challenge is that the same earthquake carries different public identifiers across agencies (a New Zealand event may appear as a GeoNet Public ID, a USGS event code, and an ISC event number), so researchers aggregating sources must de-duplicate by hand. The framework includes an event-identifier mapping service that maintains cross-agency equivalences, matched using configurable spatial, temporal, and magnitude thresholds (illustratively, separation < 20 km, time offset < 30 s, magnitude difference < 0.5). For comparison, the ANSS composite catalogue identifies duplicates on space and time alone, using operational windows of 16 s and 100 km (Northern California Earthquake Data Center, 2024); the magnitude criterion is added here to reduce false matches in dense sequences. These default thresholds will be calibrated during the pilot phase against labelled New Zealand event pairs, dense aftershock sequences, and cross-agency catalogues to estimate false-match and missed-match rates. Because wide windows can group genuinely distinct events in dense sequences, thresholds are configurable per source and ambiguous matches are flagged for expert review rather than merged automatically. The service aligns with the approach used by EMSC (European-Mediterranean Seismological Centre, 2025b) and extends it to the New Zealand context.

4.4 Catalogue Merge and Conflation

Catalogue conflation, combining events from multiple independent catalogues into a unified view, is a central operation that builds on established cross-agency practice (Si et al., 2024; Northern California Earthquake Data Center, 2024).[1] The framework specifies five merge strategies, all implemented in the prototype (Section 3.1); users choose by scientific objective:

- **Quality-based** (default): the highest quality-scored solution (Section 4.5) is retained.
- **Priority-based**: a nominated authoritative agency's solution is preferred; others are recorded as alternatives.
- **Newest data**: the most recently determined solution is retained, where later analyses supersede earlier ones.
- **Most complete**: the solution with the most populated fields is retained, maximising metadata density.
- **Averaged**: location, depth, and magnitude are combined across solutions by inverse-variance weighting, using an ISC magnitude-type hierarchy to avoid scale saturation. Agency solutions for one event often share data and processing assumptions, so this strategy is best treated as exploratory unless validated for the catalogues being combined. Its merged uncertainty is reported conservatively (no smaller than the smallest contributing value) rather than the tighter value that would follow from assuming the solutions are statistically independent.

[1] A companion review of catalogue-merging strategies accompanies this paper in the project repository (see Data and Code Availability).

For the averaged strategy, a quantity x reported by sources i with standard deviations σ_i is combined as the inverse-variance-weighted mean, alongside the variance that would follow *if* the solutions were statistically independent:

$$\bar{x} = \frac{\sum_i w_i\, x_i}{\sum_i w_i}, \qquad w_i = \frac{1}{\sigma_i^2}, \qquad \sigma_{\mathrm{indep}}^2 = \frac{1}{\sum_i w_i}.$$

The weights w_i pull the mean toward the better-constrained solution. However, agency solutions for the same event typically share waveforms, phase picks, and processing assumptions, so they are correlated rather than independent, and σ_{indep} understates the true uncertainty. The framework therefore reports a conservative uncertainty floored at the most precise contributing solution,

$$\sigma_{\mathrm{reported}} = \max\bigl(\sigma_{\mathrm{indep}},\ \min_i \sigma_i\bigr).$$

Conflicts are detected and logged when solutions disagree beyond configurable thresholds (spatial separation, temporal offset, magnitude difference, depth range, network mismatch); duplicate detection during dense aftershock sequences follows nearest-neighbour methods (Vorobieva et al., 2022). Each merged event retains explicit lineage to its source events and the configuration used, enabling full audit and reproducibility; a dry-run preview returns conflict statistics without committing a new version. Production merge configurations should report validation results and sensitivity to threshold choice. Table 5 works this through for one event reported by two catalogues, and Figure 3 shows the resulting epicentre.

Table 5: Synthetic worked example of the *averaged* merge strategy for one earthquake reported by two catalogues (Source A is entry `catA-0001`, Source B is `catB-0172`; the merged event `merged-0001` retains lineage to both). Each quantity is the inverse-variance-weighted mean of the two solutions; the normalised weights $w_A : w_B$ (with $w_i = 1/\sigma_i^2$) show how the mean is pulled toward the better-constrained Source A. The last column contrasts the uncertainty obtained by *incorrectly* assuming the solutions are independent, σ_{indep}, with the conservative value actually reported, $\sigma_{\mathrm{reported}} = \max(\sigma_{\mathrm{indep}}, \min_i \sigma_i)$ (Section 4.4); the floor is active in every row. Latitude and longitude are weighted by the horizontal uncertainty, so their merged epicentre uncertainty is the *Horizontal unc.* row. Values and identifiers are illustrative and are not official event solutions.

Quantity	**Source A**	**Source B**	**Weights** $w_A : w_B$	**Merged mean**	**Merged** σ **(indep. → rep.)**
Origin time (s)	12.3 ± 0.5	12.7 ± 0.8	0.72 : 0.28	12.4	$0.42 \rightarrow 0.50$
Latitude (°)	−38.520	−38.524	0.80 : 0.20	−38.521	n/a
Longitude (°)	176.120	176.150	0.80 : 0.20	176.126	n/a
Horizontal unc. (km)	1.5	3.0	n/a	n/a	$1.34 \rightarrow 1.50$
Depth (km)	8.0 ± 2.0	9.0 ± 4.0	0.80 : 0.20	8.2	$1.79 \rightarrow 2.00$
Magnitude (M_w)	4.2 ± 0.1	4.3 ± 0.2	0.80 : 0.20	4.22	$0.089 \rightarrow 0.10$

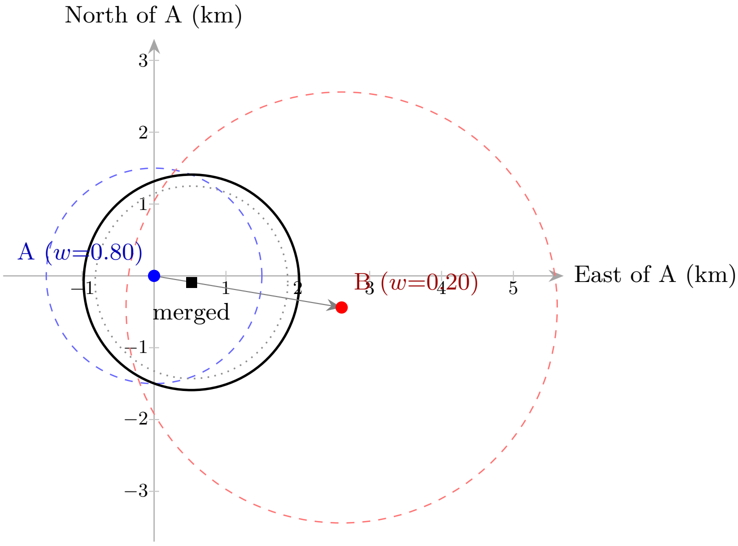

Figure 3: The averaged merge of Table 5, in a local East–North frame centred on Source A. The inverse-variance weights (0.80 : 0.20 horizontally) pull the merged epicentre (black square) about one-fifth of the way from the better-constrained Source A (blue) toward Source B (red); the dashed circles are the 1σ horizontal uncertainties (1.5 and 3.0 km). Crucially, the merged uncertainty is *not* allowed to shrink below the most precise input: the solid circle is the reported $\sigma = 1.5$ km, whereas the dotted circle is the smaller $\sigma_{\text{indep}} = 1.34$ km that the (invalid) independence assumption would give. Epicentre coordinates are plotted from the unrounded weighted mean.

4.5 Data Quality Assessment

A standardised, transparent quality score can support informed reuse decisions, building on established practice for assessing catalogue quality (Woessner and Wiemer, 2005) and on the criteria used to benchmark GeoNet location quality (Warren-Smith et al., 2025). Each event receives a score (0–100) computed as a weighted sum over five dimensions of solution quality:

- **Location** (35 %): horizontal, depth, and time uncertainties, weighted most heavily on horizontal uncertainty;
- **Network geometry** (25 %): azimuthal gap and used-station and used-phase counts, the primary instrumental predictors of location error (Bondár et al., 2004; Warren-Smith et al., 2025);
- **Solution fit** (15 %): the RMS arrival-time residual of the location;
- **Magnitude** (15 %): magnitude uncertainty and the number of contributing stations; and
- **Evaluation** (10 %): manually reviewed/final solutions score above automatic/preliminary ones.

Each dimension penalises missing fields, so sparsely documented events are scored conservatively. Field completeness, physical-consistency, and magnitude-homogeneity checks are handled separately by the automated validation suite (Appendix B). The score maps to the grade bands in Table 6. The weights and bands are configurable defaults; the pilot phase will calibrate them against representative New Zealand catalogues. The grades are screening guidance based on documentation and solution indicators, not a measure of event accuracy or of fitness for a specific hazard, engineering, or scientific analysis. Flagged statistical outliers are annotated for caution rather than removed automatically.

Table 6: Quality grade bands and screening guidance (default scheme; the pilot phase will calibrate the weights and bands; full eight-band scheme in Appendix B).

Grade	Score	Interpretation and screening guidance
A	85–100	Strong solution indicators; still verify suitability for the intended analysis.
B	65–84	Usable for screening; document limitations or supplement metadata before quantitative use.
C	45–64	Weak indicators; expert review recommended before quantitative use.
D/F	<45	Minimal documented support; discovery or illustrative purposes only.

4.6 Versioning, Persistent Identifiers, and Provenance

Every catalogue release carries a version, timestamp, and changelog, and all versions remain accessible for reproducibility. The framework uses a three-level MAJOR.MINOR.PATCH scheme (worked examples in Table 11, Appendix B): *major* increments alter event parameters, *minor* increments add events or optional fields, and *patch* increments change only metadata or documentation.

Persistent identifiers make those releases citable. The catalogue series carries a single concept DOI, and each major and minor release (the changes that affect citable content) receives its own version DOI (DataCite, 2025); patch releases update the existing record without a new DOI, and frequent minor releases may be batched (for example, quarterly) to avoid identifier proliferation. A machine-readable changelog accompanies every release, so a researcher can cite the exact data state used in a study and a reviewer can see what changed between versions.

Processing steps (detection, picking, location, velocity models, magnitude calibration) and provenance are documented following the W3C PROV model of entities, activities, and agents, with software version and execution date as minimum requirements (World Wide Web Consortium, 2013). Uncertainties for time, location, and magnitude are reported as standard deviations (with confidence-ellipse parameters where anisotropic) and always carry the magnitude type code; events without quantified uncertainties are flagged, and historical events may carry a qualitative confidence class (e.g. "well-constrained", "felt-only"). Field-level representations are specified in Appendix B.

4.7 Special Catalogue Types

Different catalogue types need adapted handling to preserve scientific value:

- **Regional research catalogues** are ingested as distinct, separately versioned catalogues and cross-referenced, not silently merged, with the national catalogue, because different processing assumptions can introduce systematic biases.
- **Temporary deployments** are labelled with their time window and station geometry.
- **Historical and pre-instrumental catalogues** use qualitative confidence classes and macroseismic intensity fields (Modified Mercalli or EMS-98, Grünthal, 1998), and form a distinct collection.
- **Derived and synthetic catalogues** carry explicit provenance to their source versions.

Appendix B gives the detailed handling.

4.8 Licensing and Access; FAIR Compliance Indicators

To support FAIR reusability, every dataset and metadata record should carry an explicit, machine-readable licence (Wilkinson et al., 2016; GO FAIR Initiative, 2025). For government-funded baseline products, a default open licence is recommended, aligned with New Zealand Government Open Access and Licensing (NZGOAL) guidance (New Zealand Government, 2022), with documented exceptions for embargoed, sensitive, or culturally restricted data. Access is governed by a role-based model with catalogue-level public/registered/restricted designations and, where needed, field-level masking of culturally sensitive locations; all write operations are audited. These mechanisms are detailed in Appendix B. Progress should be tracked against a FAIR compliance matrix that maps each FAIR sub-principle (F1–F4, A1–A2, I1–I3, R1.1–R1.3) to a framework mechanism and a measurable, annually reported indicator: for example, the percentage of catalogues with a resolvable DOI (F1), mean metadata-completeness (F2), API uptime (A1), and the percentage carrying a machine-readable licence (R1.1). The full matrix is given in Table 14 (Appendix B).

4.9 Aotearoa New Zealand Data Governance: FAIR with CARE

In Aotearoa New Zealand, openness operates within Te Tiriti o Waitangi: the Crown–Māori partnership creates obligations of governance, protection, and active participation that extend to data about, or of significance to, tangata whenua. Earth Sciences New Zealand is a Crown research entity, so the framework must give effect to those obligations, applying the CARE Principles for Indigenous Data Governance (Carroll et al., 2020) alongside FAIR and consistent with Māori data-sovereignty practice (Kukutai and Taylor, 2016).

Concretely, the four CARE principles are operationalised as follows. *Collective benefit*: engagement and benefit-sharing arrangements for catalogues of significance to iwi and hapū. *Authority to control*: iwi and hapū determine the access and use conditions for their data; the portal's access controls (Appendix B) only *implement* conditions that iwi and hapū set, and never substitute for that authority. *Responsibility*: documented engagement pathways, attribution, and reporting back to communities. *Ethics*: review of culturally sensitive records before release. Here "mātauranga-informed" means catalogue records or interpretations that draw on Māori knowledge, for example, oral-tradition accounts of historical earthquakes or place-based knowledge of shaking and ground effects, which carry their own provenance and access expectations.

For this commitment to be substantive rather than aspirational, iwi and hapū must be engaged as partners and decision-makers from the outset, and the Steering Committee (Section 5.1.1) must include Māori membership with a substantive governance role, not a single advisory seat. Phase 1 will define, with iwi and hapū partners, how culturally significant records are identified, who has authority to set access conditions, how representatives are selected and resourced, and how disputes over access, attribution, or release are resolved. The CARE implementation and access-control model will be co-developed and reviewed with mandated Māori data-governance partners.

5 Implementation: Governance, Roadmap, and Funding

5.1 Governance, Roles, and Responsibilities

5.1.1 Steering Committee

A Steering Committee provides strategic oversight and ensures community ownership. Membership should include Earth Sciences New Zealand, at least two universities with active seismology programmes, a government user (e.g. the National Emergency Management Agency), an engineering or insurance representative, Māori representation with the mandate and resourcing to speak to data-sovereignty matters (see Section 4.9), and, where feasible, an international advisor from a peer organisation (e.g. ISC, EMSC, or the GFZ Helmholtz Centre for Geosciences). The process for appointing Māori members should be developed with iwi and hapū partners rather than set unilaterally by the host institution. It meets at least quarterly with defined decision-making processes and staggered three-year terms. The Committee approves new data sources for ingestion, ratifies versioning decisions affecting the national record, oversees international partnerships (e.g. alignment with EPOS), and commissions annual reporting against FAIR compliance indicators.

5.1.2 Data Stewardship Team

Day-to-day operation is led by Earth Sciences New Zealand and need not be large. At its simplest the framework is maintained by a part-time data steward (coordination, metadata curation, and quality review); software-engineering and user-support effort are drawn in as needed, and the team grows toward one to two FTE only if the full services are operated at scale. External researchers may also submit catalogues through a defined contribution process (Section 5.1.3). These governance roles (contributor, reviewer, maintainer) are distinct from the portal's technical access roles defined in Appendix B.

5.1.3 Catalogue Contribution Process

External researchers and institutions may submit catalogues for inclusion. Submissions require: catalogue name and description, responsible institution and contact, time period and geographic coverage, event count where known, format and metadata documentation, and a declared licence; magnitude range is required where instrumental magnitudes are part of the catalogue and otherwise recorded as not applicable. A two-stage review process applies: technical QC by the Data Stewardship Team (format validity, completeness score, duplicate check) followed by scientific peer review by a standing panel of subject-matter seismologists who are independent of the submitted catalogue where practical (methods assessment, comparison with existing catalogues). Because the Committee meets only quarterly, responsibility is divided to keep the contribution turnaround within a target of four to eight weeks: the Steering Committee sets ingestion policy and approves *categories* of source, while the Team approves individual contributions against that policy. Only novel or contentious cases are escalated to the Committee. Written feedback is provided regardless of outcome, and approved catalogues are assigned a DOI and version record.

5.1.4 Catalogue Custody, Updates, and Use Conditions

Each catalogue entry names a custodian (the person responsible for it) with a listed contact for updates, corrections, or withdrawal. The custodian need not be the original author, so that past studies and legacy datasets can be entered by others; in such cases the entry clearly attributes the catalogue's authorship. Where the same catalogue might be entered more than once (for example, a legacy dataset entered independently), duplicate entries are identified and reconciled by the Data Stewardship Team. The catalogue index itself is version-controlled, with its release history maintained in a public Git repository, so that additions, edits, and deprecations are transparent and a superseding catalogue version can be linked to the entry it replaces.

The catalogue index is a finding aid, not an endorsement (the submission profile itself is defined in Section 4.2). A clear disclaimer states that entries are contributed and not independently verified, and that the index simply records which catalogues exist. Users are encouraged to reference catalogues appropriately and to contact and collaborate with the original authors where relevant, consistent with the intellectual-property and data-sharing expectations attached to each dataset.

5.1.5 Success Metrics and Annual Reporting

Progress should be measured annually against indicators including: number of catalogues listed and ingested; total event count and geographic coverage; metadata-completeness distributions; API uptime (target 99.5%); registered users; downloads and API queries; peer-reviewed publications citing the framework or a versioned catalogue DOI; and scientific-validity indicators such as merge precision/recall on benchmark pairs, reproducibility checks on cited versions, and user-reported time or error reductions in cross-catalogue workflows. Reporting against the FAIR sub-principle indicators (Table 14) should also be included, and annual reports published openly. Illustrative two-year targets: the GeoNet national catalogue plus 20 or more additional regional, temporary, and historical catalogues discoverable through the framework, with priority datasets ingested for the full services; 200 or more registered users across the research, engineering, and agency communities; 10 or more research outputs citing a versioned catalogue DOI; and API availability approaching the 99.5% operational target. Phase 1 will set the baselines and measurement methods against which these targets are tracked.

5.2 Roadmap

Development proceeds in four phases (Figure 4): planning and mobilisation; prototyping and pilot integration; full-scale implementation and public launch; and sustained operation as a national core service with annual review and FAIR compliance reporting.

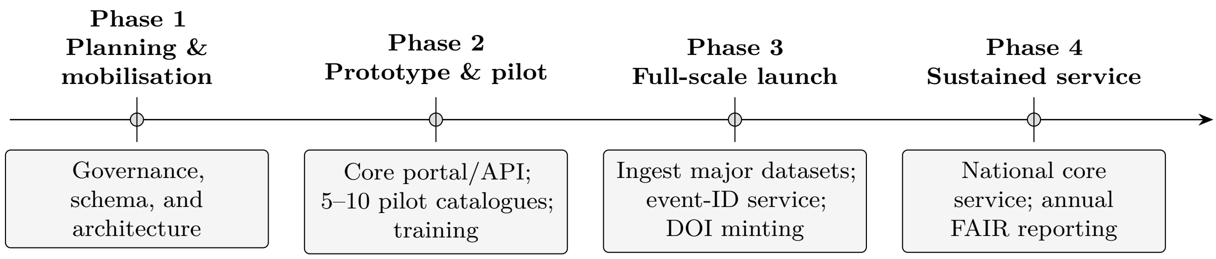

Figure 4: Indicative four-phase roadmap from planning to a sustained national service.

5.3 Funding and Resourcing

Indicative resourcing is modest, and the discovery function in particular is inexpensive to run. It can be launched and maintained by a part-time data-stewardship role (coordination, metadata curation, and periodic review) with low-cost hosting for a versioned index (Section 5.1.2), on the order of a few tens of thousands of dollars per year. A mature service that also operates the full curation and ingestion services would scale to perhaps one to two FTE, adding software-engineering and user-support effort, together with modest cloud infrastructure and recurring DOI-registration fees. A realistic path therefore begins small, with discovery, and grows only as the full services are built out and demand justifies it. Indefinite retention of every versioned release is a growing-storage liability; it is bounded by keeping only changelogs and metadata for superseded *patch* releases, retaining full event tables only for DOI-bearing major and minor versions, and moving older releases to tiered/cold storage. A natural home is a national research-infrastructure line (for example, an Earth Sciences New Zealand core-function budget or a research-infrastructure programme such as those administered by the Ministry of Business, Innovation and Employment, MBIE), so that the service is funded as a standing capability rather than a fixed-term project. The figures here are indicative of scale, not costed estimates; precise budgets should be developed with the host institution.

6 Benefits by Stakeholder

A coordinated, FAIR-and-CARE catalogue capability delivers concrete, audience-specific value:

- **Researchers** gain discoverable, citable, version-pinned datasets and ready cross-catalogue comparison, making studies faster to run and easier to reproduce.
- **Hazard and risk modellers** (e.g. the National Seismic Hazard Model) get comparable, documented inputs with consistent magnitudes and uncertainties, reducing the bespoke harmonisation that the 2022 NSHM required (Gerstenberger et al., 2024; Christophersen et al., 2024).
- **Engineering and insurance** users obtain traceable inputs with explicit licences and transparent quality scores, supporting more defensible design and risk decisions.
- **Emergency management and the public** benefit from clearer, better-documented event information and a single national point of discovery, while GeoNet remains the authoritative operational source.
- **Iwi and hapū** should be engaged as partners in governance, setting the access and attribution conditions for data of significance to them, which the framework then implements (Section 4.9).
- **The national system** preserves at-risk legacy and temporary datasets and aligns New Zealand with international best practice (EPOS, 2025b).
- **Community infrastructure** gains a stepping-stone, in the shared catalogue of catalogues, toward broader community goals identified at the workshops: digitising and back-filling legacy catalogues, and converging on a national community velocity model (1D and 3D). The framework should accept routine updates, with its governance and schema reviewed on a regular cadence.

7 Risks and Anticipated Questions

We anticipate, and address, the main objections to this proposal:

- **“Isn’t this duplicating GeoNet?”** No. The framework is an aggregation and curation layer over GeoNet and the many catalogues outside it; GeoNet remains the authoritative operational source and a primary data provider (Figure 2).

- **“Who pays once the establishment funding ends?”** The service is designed as a standing national capability hosted on a core-function or research-infrastructure budget, not a project that ends with a grant. The indicative steady-state cost and a bounded-retention storage policy are given in Section 5 (Funding).
- **“What if researchers don’t contribute their catalogues?”** The contribution process is low-friction and offers tangible incentives (DOIs, citations, increased visibility, and quality feedback), and the system delivers value from GeoNet and pilot catalogues from day one.
- **“Why not just use EPOS, ISC, or ComCat?”** Those systems are complementary and the framework is designed to interoperate with them, but none curates New Zealand’s regional, temporary, and historical catalogues or addresses CARE obligations; a national capability is still required.
- **“Will standardisation flatten legitimate scientific differences?”** No. Alternative solutions are preserved with lineage, regional catalogues are cross-referenced rather than silently merged, and merge strategies are explicit and reproducible.

8 Conclusion and Call to Action

New Zealand has the data, the expertise, and strong community interest in making its earthquake catalogues FAIR; what is missing is a coordinating framework and the governance to sustain it. This white paper has set out such a framework, compared it with international practice, and demonstrated its technical feasibility through a working prototype. In doing so it has characterised the catalogue landscape and compared it internationally, proposed minimum attributes, a metadata profile, and a shared lexicon, and set out a gap analysis with near-term community steps. The remaining step is collective review, partner agreement, and sustained implementation.

We invite GeoNet and Earth Sciences New Zealand, universities, government agencies, iwi and hapū partners, and the engineering and insurance sectors to:

1. review, refine, and, where appropriate, endorse the metadata, lexicon, versioning, and quality-scoring standards proposed here as a shared national baseline;
2. stand up the “catalogue of catalogues” as the immediate, low-cost first step, beginning with discovery (populated via the agreed submission profile) ahead of building out the full FAIR services;
3. commit to establishing the Steering Committee and Data Stewardship Team (Sections 5.1.1 and 5.1.2), with representation that operationalises both FAIR and CARE; and
4. support a multi-year funding case to move from prototype to a sustained national service.

We welcome feedback and expressions of interest in participating in the governance and catalogue-contribution processes described above. Comments may be directed to the corresponding author, Kenny Graham (Earth Sciences New Zealand).

Data and Code Availability

In keeping with the FAIR principles it advocates, the prototype portal source code, the metadata schema, the controlled parameter lexicon, the companion review of catalogue-merging strategies, and a companion inventory of New Zealand earthquake catalogues (1960–2026) are available under the MIT licence at `https://github.com/KennyGraham1/catalogofcatalogs` (documentation at `https://catalogofcatalogs.readthedocs.io/`); consistent with the persistent-identifier practice this paper advocates, a citable DOI for a versioned snapshot of these artefacts will be minted (e.g. via Zenodo) with the first tagged release. The earthquake catalogues used as examples are available from their respective providers: the GeoNet national catalogue (GeoNet, 2025a), USGS ComCat (USGS, 2025), and the Swiss Seismological Service (Diehl et al., 2025).

Acknowledgements

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A International Benchmark Detail

This appendix expands the summary in Section 2 and Table 2.

A.1 New Zealand (GeoNet)

GeoNet maintains New Zealand's earthquake catalogue as part of the National Earthquake Information Database, the central repository for NZ seismic events (GNS Science, 2020). The catalogue contains origin time, location, magnitude, and seismic phase data, with each event assigned a unique Public ID (GeoNet, 2025a), spanning instrumentally recorded earthquakes (1930s–present) and historical events from oral/written records (GNS Science, 2020). Data access includes Quake Search (CSV, GeoJSON, KML, GML) (GeoNet, 2025a) and services compliant with the International Federation of Digital Seismograph Networks (FDSN) and Web Feature Service (WFS) standards, delivering QuakeML with uncertainties, phase arrivals, and multiple magnitude determinations (GeoNet, 2025c,b). Completeness is approximately M_L 1–2 in well-monitored regions (higher elsewhere and through time); location quality varies markedly with station coverage, as quantified by Warren-Smith et al. (2025). A 2022 revision moved the catalogue toward more consistent magnitude reporting nationwide (GNS Science, 2020).

A.2 United States (USGS ComCat)

The United States Geological Survey (USGS) operates the federated Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat), covering global earthquakes ($\sim M4.5$ and above as a common global analysis threshold, and roughly $M2.5$ in well-monitored US regions) (USGS, 2025), accessed via the FDSN Event API with QuakeML, CSV/GeoJSON/KML, and real-time GeoJSON feeds (U.S. Geological Survey, 2025). Entries include quality metrics (stations used, azimuthal gap, RMS residuals) and magnitude uncertainty; ComCat integrates the ANSS regional and national networks, stores multiple solutions per event with one preferred, and hosts products such as ShakeMap and crowdsourced intensity data (USGS, 2025).

A.3 Switzerland (SED)

The Swiss Seismological Service (SED) maintains a high-quality catalogue dating to 1879 (Diehl et al., 2025), with historical events to 250 AD. A dense network achieves M_c 1.0–1.5 completeness;

SED introduced updated magnitude scales (M_{Lhc}, 2020), employs 3D velocity models and relative relocation, and provides formal uncertainties via QuakeML since 2013 (Diehl et al., 2025). Data are accessible through FDSN web services, and cross-border collaboration integrates 60+ stations from France, Germany, Italy, and Austria (Diehl et al., 2025). The ECOS project unified historical records into a homogenised database (Fäh et al., 2011).

A.4 European and Global Aggregators

The European-Mediterranean Seismological Centre (EMSC) operates the European Seismic Portal, aggregating data from multiple networks via FDSN-compliant services and providing multiple solutions and phase data (European-Mediterranean Seismological Centre, 2025a,b). The International Seismological Centre (ISC) archives global bulletin data from 130+ agencies, accepting multiple formats (Nordic, GSE, ISF, QuakeML) (International Seismological Centre, 2025a) with output in ISF or QuakeML (International Seismological Centre, 2025c). Other agencies (the Japan Meteorological Agency, Geoscience Australia, and the European Integrated Data Archive) have adopted FDSN services, and major catalogues now share common standards for open, interoperable access (EPOS, 2025b; ETH Zürich Seismology / QuakeML Working Group, 2015).

B Technical Specification

This appendix collects the implementation-level detail referenced from Section 4. It defines the reference design; specific field representations and thresholds may be refined with partners during Phase 1.

B.1 Metadata Field Hierarchy

Table 7 separates requirement levels by catalogue scope, and Table 8 gives the catalogue- and event-level metadata profile for ingested records.

Table 7: Metadata requirement levels across the framework.

Scope	**Requirement level**	**Purpose**
Discovery entry (submission profile)	Minimal descriptor required	Discovery, attribution, contact, coverage, and availability; data may remain external or unavailable.
Ingested catalogue	Event-level structure required where applicable	Enables validation, search, analytics, exports, and event-ID mapping.
DOI-bearing release	Version, changelog, licence, and persistent identifier required	Ensures a citable, reproducible data state.
Legacy or historical record	Missing quantified fields must be flagged	Preserves discoverability while avoiding false precision.

Table 8: Metadata Field Hierarchy for Ingested Catalogue Records

Field	Status	Description
Catalogue-level		
Persistent identifier	Required for DOI-bearing releases	DOI or equivalent resolvable identifier; discovery-only entries may instead point to an external contact or source
Name, description	Required	Human-readable title and summary
Institution, contact	Required	Responsible party and contact details
Temporal coverage	Required	Start and end date of events
Geographic coverage	Required	Bounding box or region descriptor
Magnitude range	Required where applicable	Minimum and maximum magnitude; record as not applicable for catalogues without instrumental magnitudes
Event count	Required where known; required for ingested catalogues	Total number of events
Licence	Required	Machine-readable licence identifier (e.g. CC-BY 4.0)
Processing summary	Required	High-level description of methods
Velocity model	Recommended	Reference to velocity model used
Location algorithm	Recommended	Software and version (e.g. NonLinLoc 7.0)
Magnitude calibration	Recommended	Scale definitions and calibration reference
Publication links	Recommended	DOIs of associated papers
Event-level		
Origin time	Required	UTC time (ISO 8601)
Latitude, longitude	Required	WGS84 decimal degrees
Depth	Required	Kilometres below sea level
Magnitude value & type	Required where applicable	Value and scale code (e.g. M_w, M_L); historical/intensity-only records may instead carry intensity or qualitative size fields
Origin time uncertainty	Required where available	Seconds (1σ); otherwise flag absence explicitly
Horizontal uncertainty	Required where available	Kilometres (1σ); otherwise flag absence explicitly
Vertical uncertainty	Required where available	Kilometres (1σ); otherwise flag absence explicitly
Magnitude uncertainty	Required where available	Standard deviation; otherwise flag absence explicitly
Station count	Recommended	Number of stations used
Azimuthal gap	Recommended	Degrees
Phase count	Recommended	Number of phase picks used
RMS residual	Recommended	Seconds
Evaluation status	Recommended	“automatic” or “reviewed”
Event type	Recommended	QuakeML event type vocabulary

B.2 Controlled Parameter Lexicon

Table 9 shows representative entries from the controlled lexicon referenced in Section 4. It is an illustrative subset, aligned with the QuakeML enumerations (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016), not the complete normative list. A fuller controlled vocabulary is maintained with the schema in the project repository (see Data and Code Availability); its normative, versioned form will be published with the first DOI-bearing release.

Table 9: Illustrative subset of the controlled parameter lexicon (representative terms; aligned with QuakeML).

Category	**Representative controlled terms**
Magnitude type	Mw (moment); ML, MLv (local); mb (body-wave); Ms (surface-wave); Md (duration)
Event type	earthquake; induced or triggered event; quarry blast; explosion; volcanic eruption; other event
Phase name	P, S; Pn, Pg, Sn, Sg (crustal); pP, sP (depth phases)
Evaluation mode / status	mode: manual / automatic; status: preliminary / confirmed / reviewed / final / rejected
Pick onset / polarity	onset: emergent / impulsive / questionable; polarity: positive / negative / undecidable
Depth determination	from location; operator assigned; constrained by depth phases; constrained by direct phases
Units & formats	time: ISO 8601 UTC; coordinates: WGS84 decimal degrees; depth: km below sea level

B.3 QuakeML/FDSN Crosswalk

The framework adopts QuakeML 1.2 and the FDSN web-service conventions as its interchange backbone (ETH Zürich Seismology / QuakeML Working Group, 2015). Table 10 maps the framework's catalogue constructs to specific QuakeML elements. Most map directly to native QuakeML. QuakeML also permits free-text `comment` elements and custom elements from a foreign namespace; the framework uses these, together with companion records (W3C PROV provenance (World Wide Web Consortium, 2013), the DataCite DOI record, and the catalogue submission profile, Table 4), to carry the constructs that have no native QuakeML element: chiefly the composite quality score, cross-catalogue merge lineage, qualitative confidence classes, and macroseismic intensity for historical events. A complete, versioned crosswalk will be published with the schema (see Data and Code Availability) and finalised in Phase 1.

Table 10: Crosswalk from framework catalogue constructs to QuakeML 1.2 / FDSN. Status: native = direct QuakeML element; ext. = QuakeML `comment` or foreign-namespace extension; comp. = carried in a companion record (W3C PROV, DataCite DOI, or the submission profile).

Framework construct	**QuakeML 1.2 / FDSN representation**	**Status**
Event and solution identifiers	`event@publicID`, `origin@publicID`, `magnitude@publicID`	native
Origin time and hypocentre	`origin/time`, `/latitude`, `/longitude`, `/depth` (each `value` with `uncertainty`)	native
Location uncertainty / confidence ellipse	`origin/originUncertainty` (`horizontalUncertainty`, `minHorizontalUncertainty`, `maxHorizontalUncertainty`, `azimuthMaxHorizontalUncertainty`, `confidenceEllipsoid`, `preferredDescription`)	native
Network geometry and fit	`origin/quality` (`azimuthalGap`, `usedStationCount`, `usedPhaseCount`, `standardError`)	native
Depth type; method; earth model	`origin/depthType`, `origin/methodID`, `origin/earthModelID`	native
Evaluation mode and status	`origin/evaluationMode`, `origin/evaluationStatus`	native
Magnitude (value, type, error, stations)	`magnitude/mag` (`value`, `uncertainty`), `magnitude/type`, `magnitude/stationCount`	native
Phase picks and arrivals	`pick` (`time`, `phaseHint`, `onset`, `polarity`); `arrival` (`phase`, `azimuth`, `distance`, `timeResidual`)	native
Event type and classification	`event/type`, `event/typeCertainty`, `description`	native
Macroseismic intensity (historical)	No native element in QuakeML 1.2 (macroseismic is a QuakeML 2.0 package, still under development); carried via foreign-namespace extension / `comment`; intensity recorded as MMI or EMS-98	ext.
Creation provenance	`creationInfo` (`agencyID`, `author`, `version`, `creationTime`)	native
Composite quality score (0–100) and grade	No native element; framework-namespace extension on `origin`, derived from the `origin/quality` inputs above	ext.
Qualitative confidence class; uncertainty-absent flag	Controlled `comment` / framework extension; `uncertainty_available` = false	ext.
Merge lineage / source-event mapping ("derivedFrom")	No native construct; W3C PROV-O records (World Wide Web Consortium, 2013) and the event-identifier mapping service, referenced from `comment`	comp.
Catalogue-level descriptors (submission profile)	Minimal native support (`eventParameters/description`, `comment`); carried in the submission profile (Table 4)	comp.
Persistent identifier and dataset version	DataCite DOI at dataset level; `creationInfo/version` per record	comp.

B.4 Versioning Scheme

Table 11 illustrates the version-increment triggers for the three-level MAJOR.MINOR.PATCH scheme described in Section 4.

Table 11: Versioning Scheme: Examples of Version Increment Triggers

Version	Trigger	Example change
MAJOR	Event parameters changed	Reprocessing with new 3D velocity model; recalibration of magnitude scale; new location algorithm affecting systematic offsets
MINOR	New events or fields added	Ingestion of 500 additional aftershock events; addition of focal mechanism fields to existing events
PATCH	Metadata or documentation	Contact email updated; description clarified; deprecated field removed from catalogue-level record

B.5 API Design

A unified API follows REST conventions. Authentication uses token-based credentials for write operations; read operations on public catalogues are unauthenticated. Rate limiting protects service availability, and standard HTTP error codes with a consistent JSON error envelope enable reliable programmatic error handling. Output format is negotiated via the `Accept` header or a `format` query parameter, defaulting to GeoJSON for spatial queries and CSV for tabular exports; QuakeML output is generated from the canonical internal event model, preserving uncertainty fields, evaluation status, and magnitude type. Table 12 summarises the principal endpoint categories.

Table 12: Principal API Endpoint Categories

Category	Operations	Notes
Catalogues	List, retrieve, create, update, delete	Metadata including bounds, event count, licence
Events	List, spatial query, time-range, magnitude/depth filter	Pagination; supports GeoJSON, CSV, QuakeML output
Search	Cross-catalogue full-text and parameter search	Returns ranked results across all accessible catalogues
Import	Upload file, sync from FDSN source, view history	Supports CSV, JSON, GeoJSON, QuakeML 1.2
Merge	Preview, execute, retrieve conflict log	Dry-run mode returns statistics without committing
Export	Per-catalogue or per-query download	CSV, GeoJSON, KML, QuakeML; version-specific
Versions	List versions, retrieve specific version	Version-specific event snapshots remain available indefinitely

B.6 Access Control and Audit

Access to catalogue data and portal operations is governed by a role-based model. Four *technical* roles are defined (distinct from the governance roles of Section 5.1.3; a single individual may hold both):

- **Administrator**: full access including user management, system configuration, and deletion of catalogue records.
- **Editor**: can create and modify catalogues, import data, and initiate merge operations, but cannot manage users or alter system configuration.
- **Viewer**: can search, view, filter, and export catalogues to which they have been granted access, but cannot modify data.
- **Guest** (unauthenticated): can discover and download public catalogues; embargoed or restricted catalogues are not visible.

Role assignments are managed by Administrators and recorded in an audit log; a role-upgrade request workflow allows registered users to request elevated access.

Data-level access control. Each catalogue carries one of three access designations: **public** (all, unauthenticated), **registered** (authenticated Viewers and above), or **restricted** (named individuals or groups). This supports time-limited embargoes (e.g. allowing a research group to work privately until a companion paper is published). Field-level masking may be applied where a catalogue contains sensitive location information, such as event locations near culturally significant sites: coordinates may be spatially generalised or withheld for Guest and Viewer access while full precision is retained for authorised users, consistent with the CARE discussion in Section 4.9.

Audit and compliance. All write operations (catalogue creation, import, merge, deletion) are recorded with the authenticated user identity, timestamp, and operation details; read access to restricted catalogues is similarly logged. Where usage agreements are attached to a catalogue (for example, a requirement to cite a companion publication), the agreement is presented at download time and acceptance is recorded.

B.7 Uncertainty Representation

Origin location and time uncertainties are reported as standard deviations where available: `time_uncertainty` (seconds), `horizontal_uncertainty` (kilometres), and `depth_uncertainty` (kilometres). Where horizontal uncertainty is anisotropic, the semi-major and semi-minor axes of the confidence ellipse, its azimuth, and the stated confidence level (conventionally 68%) are reported, mapping to QuakeML `OriginUncertainty` elements (ETH Zürich Seismology / QuakeML Working Group, 2015). Magnitude uncertainties are reported as the standard deviation on the stated scale when available, and the magnitude type code must always accompany the value. Events without quantified uncertainties carry a boolean `uncertainty_available` field set to false; historical events may substitute a qualitative confidence class. Thus the framework requires uncertainty to be represented explicitly, but does not require every legacy or historical record to have a quantified uncertainty. When events are merged, the averaging strategy combines location, time, and magnitude by inverse-variance weighting; because agency solutions for one event are typically correlated rather than independent, the merged uncertainty is reported conservatively (not below the smallest contributing value) rather than the smaller independent-combination result. Priority and quality strategies retain the selected solution's uncertainty and record alternatives; the merge log documents the strategy and resulting uncertainty.

B.8 Quality Checks and Full Grade Scheme

Representative automated checks include:

- events ordered chronologically, with no future-dated events;
- depth within a physically plausible range for the region (for New Zealand, about -5 to 700 km, covering Hikurangi–Kermadec subduction depths and allowing for events above the sea-level datum; a higher default applies to global catalogues);
- magnitude within a plausible range for the declared region;
- uncertainty values non-negative and within parameter-specific sanity bounds (for example, horizontal uncertainty not exceeding the catalogue's declared spatial extent, and magnitude uncertainty below a configurable ceiling such as 1.0 unit);
- no coordinate pair repeated for distinct events at an identical time (duplicate detection);
- station count ≥ 5 for an event to be considered well-constrained; and
- azimuthal gap $< 180°$ for a reliably constrained epicentral location (USGS, 2025).

Results are summarised as passed, warning, or failed. These checks are sanity screens and documentation indicators, not accuracy tests. The full eight-band grade scheme underlying the summary in Table 6 is given in Table 13.

Table 13: Full quality grade thresholds and screening guidance (default scheme; the pilot phase will calibrate the thresholds).

Grade	Score	Interpretation and screening guidance
A+	95–100	Very strong documented indicators; still verify suitability for the intended analysis.
A	85–94	Strong documented indicators; suitable for screening and candidate selection.
B+	75–84	Moderate indicators; document limitations in methods sections.
B	65–74	Fair indicators; consider supplementing with additional metadata.
C+	55–64	Weak indicators; expert review recommended before quantitative use.
C	45–54	Very weak indicators; limited utility for quantitative analysis without review.
D	35–44	Minimal documented support; use only for discovery or illustrative purposes.
F	<35	Not recommended for quantitative analysis without substantial review.

B.9 Special Catalogue Type Handling

Regional research catalogues produced with specialised methods (e.g. 3D velocity models, relative relocation, custom magnitude calibrations) are ingested as distinct catalogues with their own version records and DOIs; processing methods are documented in catalogue-level metadata, and the portal provides cross-reference links between co-temporal events in regional and national catalogues with the spatial and magnitude offset reported. **Temporary deployment catalogues** are ingested with an explicit time window and a "research" or "monitoring" designation, with station-geometry metadata and links to relevant FDSN waveform services. **Historical and pre-instrumental catalogues** are accommodated through qualitative location confidence classes, macroseismic intensity fields (Modified Mercalli or EMS-98, Grünthal, 1998), narrative source-attribution fields, and separate quality thresholds, and are presented as a separate collection. **Derived and synthetic catalogues** carry mandatory provenance identifying source catalogue(s), derivation method, parameters, and software, with explicit "derivedFrom" relations to source versions; synthetic catalogues are labelled as such.

B.10 FAIR Compliance Matrix

Table 14 gives the full mapping summarised in Section 4: each FAIR sub-principle, the framework mechanism that addresses it, and a measurable indicator for annual reporting.

Table 14: FAIR compliance matrix: sub-principle, framework mechanism, and indicator.

Sub-pr.	**Framework mechanism**	**Indicator**
F1	Persistent identifiers (DOIs) for catalogues; stable event identifiers	% of catalogues with a resolvable DOI; % of events with a stable ID
F2	Rich metadata profile (catalogue- and event-level)	Mean metadata-completeness score across catalogues
F3	Metadata records embed the data identifier	% of metadata records linking to their data identifier
F4	Indexing in the searchable portal and discovery API	% of catalogues indexed and discoverable
A1	Retrieval via FDSN/REST over HTTPS	API uptime; % of catalogues retrievable by identifier
A1.1–2	Open, standard protocols; token auth only for write/restricted data	% of read endpoints requiring no authentication
A2	Metadata and version records retained after data withdrawal	% of withdrawn datasets with surviving metadata
I1	QuakeML / FDSN standard representations	% of catalogues available as valid QuakeML
I2	Controlled parameter lexicon (magnitude, event, phase types)	% of fields using controlled-vocabulary terms
I3	Qualified cross-references (event-ID mapping, lineage links)	% of merged events with recorded source lineage
R1.1	Machine-readable licence on every record	% of catalogues with a machine-readable licence
R1.2	W3C PROV-based provenance (entities, activities, agents)	Mean provenance-completeness score
R1.3	Conformance to FDSN/QuakeML community standards	% of catalogues passing standards-compliance checks

C Summary of Workshop Themes

This appendix summarises the workshop discussions, synthesised here as nine cross-cutting themes. The content reflects community perspectives on minimum catalogue attributes, metadata requirements, processing concerns, governance needs, and priorities for a national FAIR-aligned framework for earthquake catalogues in Aotearoa New Zealand.

Table 15: Summary of Key Themes from the Workshop Discussions

Theme	Key Points
Minimum Catalogue Attributes	Phase picks; station metadata; velocity models (1D/3D); depths and magnitudes with uncertainties; completeness estimates; region of interest; unique IDs; focal mechanisms; processing history; detection and processing methods.
Metadata & Documentation Needs	Clear methodology (location algorithm, velocity model, filters); solution provenance; waveform availability; descriptive summaries; parameter definitions; assumptions (point-source vs 2D/3D); embargo details; QC information; required README and DOI; multiple contacts.
Access & Versioning	Raw data accessibility; open formats (CSV, QuakeML, FDSN); programmatic access (API); search by region, time, magnitude; support for multiple versions; backward/forward compatible formats; tracking of method changes; long-term storage of legacy releases.
Governance Needs	Defined roles (contributor, reviewer, maintainer); central catalogue portal; consistent naming conventions; periodic updates; upload validation; framework transparency; responsibility for long-term stewardship.
Community Priorities	National catalogue portal; common parameter lexicon; national 3D community velocity model; honest uncertainties; digitisation & reprocessing of legacy data; cross-catalogue interoperability; clear documentation for older datasets.
Concerns Raised	Inconsistent completeness; variable processing; missing metadata; uncertain magnitude/depth scales; "black-box" ML outputs; compatibility issues across regional/temporary catalogues; handling of unpublished datasets; need for quality flags and user warnings.
Desired Portal Features	Merge multiple catalogues; support subcatalogs; synthetic catalogues; geographic reference tools; method-based filtering; visualisation; minimum-requirements upload form; linkage among event, waveform, and station data.
What Users Liked	Simple form; unified access; clear uncertainty fields; reproducibility focus; increased awareness of catalogue availability; prioritisation of essential fields.
Requested Improvements	More metadata fields and filters; better uncertainty representation; link with 2D/3D models; clarify naming/versioning; transparent ML integration; improved handling of old locations; inclusion of focal mechanisms and strong-motion connections.

D Glossary

ANSS / ComCat Advanced National Seismic System; its Comprehensive Earthquake Catalog (USGS).

Azimuthal gap the largest angle, seen from an epicentre, between adjacent recording stations; a smaller gap means a better-constrained location.

CARE Collective benefit, Authority to control, Responsibility, Ethics: principles for Indigenous data governance.

CMT / GCMT centroid moment tensor; GCMT is the Global CMT project's moment-tensor catalogue.

Conflation / merge combining records of the same earthquake from different catalogues into one unified event.

Declustering separating mainshock–aftershock sequences from background seismicity (e.g. the Gardner–Knopoff method).

DOI Digital Object Identifier: a persistent, citable identifier for a dataset or document.

EMSC / ISC European-Mediterranean Seismological Centre; International Seismological Centre.

EPOS European Plate Observing System: a coordinated FAIR data infrastructure for solid-Earth science.

FAIR Findable, Accessible, Interoperable, Reusable: principles for scientific data management.

FDSN International Federation of Digital Seismograph Networks; its web services are a global standard for seismic data access.

GeoNet New Zealand's geological-hazard monitoring programme, operated by Earth Sciences New Zealand.

M_c magnitude of completeness: the lowest magnitude above which all earthquakes in a catalogue are reliably recorded.

M_w**,** M_L moment magnitude and local (Richter) magnitude: two magnitude scales.

NSHM (Aotearoa New Zealand) National Seismic Hazard Model.

OBS ocean-bottom seismometer: a seafloor instrument used in temporary offshore deployments.

PROV / PROV-O the W3C provenance data model and ontology (entities, activities, and agents).

Provenance the documented origin and processing history of a dataset (inputs, methods, software, agents).

PSHA probabilistic seismic hazard analysis.

QuakeML / StationXML standard XML formats for exchanging earthquake parameters and station metadata.

WFS Web Feature Service: an OGC standard for serving geographic features.

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